

TechSpark Solutions Pvt. Ltd.

AI Products Division | Noida, Uttar Pradesh
careers@techspark.in • www.techspark.in

JOB DESCRIPTION

Machine Learning Engineer – Intern

Python	LangChain	RAG	NLP	scikit-learn	Fresher OK
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ROLE OVERVIEW

Job Title	Machine Learning Engineer – Intern
Department	AI Products & Research
Location	Noida, Uttar Pradesh (Hybrid – 3 days on-site)
Duration	6 Months (with full-time conversion possibility)
Stipend	₹15,000 – ₹20,000 per month
Start Date	Immediate / July 2025
Experience	Fresher / 0–1 year (B.Tech students in 3rd or 4th year welcome)
Openings	2

ABOUT TECHSPARK SOLUTIONS

TechSpark Solutions is a Noida-based AI product company building intelligent tools for the Indian MSME and EdTech sectors. Our team of 60+ engineers works on NLP, computer vision, and intelligent automation systems that serve over 2 lakh users daily. We believe in hiring smart, driven learners and growing them into full-stack AI engineers.

ROLE DESCRIPTION

We are looking for a passionate and hands-on Machine Learning Engineering Intern to join our AI Products team. You will work directly with senior ML engineers on live production projects including our document intelligence platform and internal chatbot system. This is a builder role — you will write real code, ship real features, and learn at 10x pace.

KEY RESPONSIBILITIES

- Assist in building and maintaining RAG (Retrieval-Augmented Generation) pipelines for document Q&A features using LangChain and FAISS.
- Implement and experiment with open-source LLMs (LLaMA, Mistral) locally and on cloud for inference tasks.
- Write clean, well-documented Python code for data preprocessing, feature engineering, and model evaluation.
- Work with vector databases (FAISS, ChromaDB) to build and optimise semantic search systems.

- Support development of REST APIs using FastAPI to serve ML model outputs to frontend teams.
- Run experiments, track results using MLflow or similar tools, and document findings clearly.
- Collaborate with the product team to understand requirements and translate them into ML solutions.
- Participate in weekly code reviews and team knowledge-sharing sessions.

REQUIRED SKILLS & QUALIFICATIONS

- Currently pursuing B.Tech / B.E. in Computer Science, IT, or related field (3rd or 4th year preferred).
- **Python** Strong proficiency in
 - NumPy, Pandas, Matplotlib for data wrangling and visualisation
 - scikit-learn for classical ML models — regression, classification, clustering
- **Machine Learning concepts:** Working knowledge of
 - Supervised / unsupervised learning, model evaluation metrics, overfitting/underfitting
 - Logistic Regression, Decision Trees, k-NN — understanding of how and when to use them
- **LangChain or LLM APIs** Familiarity with
 - Has built or experimented with at least one RAG or chatbot project (personal or academic)
- **Git & GitHub** Basic understanding of
 - Able to maintain a GitHub repo, commit consistently, and document projects clearly
- **research paper summaries and ML documentation** Ability to read and understand
- Good written and verbal communication in English.

GOOD TO HAVE (BONUS POINTS)

- Experience running local LLMs using Ollama or similar tools on consumer hardware.
- Exposure to vector databases such as FAISS, ChromaDB, or Pinecone.
- Basic knowledge of FastAPI or Flask for building ML-serving APIs.
- Familiarity with Docker and basic cloud concepts (Azure, AWS, GCP).
- Completed any online ML/AI course (Coursera, Andrew Ng's specialisation, fast.ai, etc.).
- Active GitHub profile with at least one ML project with a descriptive README.
- Interest in Agentic AI, LLM fine-tuning, or computer vision is a plus.

WHAT YOU WILL LEARN & GAIN

- End-to-end ML pipeline development — from data ingestion to model deployment.
- Real-world experience with production-grade RAG systems and LLM integration.
- MLOps basics — experiment tracking, model versioning, and monitoring.
- Exposure to Agile development, code review culture, and professional engineering practices.
- Mentorship from senior engineers and a structured learning path toward full-time ML Engineering.
- Certificate of Internship + Performance-based Pre-Placement Offer (PPO) consideration.

SELECTION PROCESS

- Round 1 – Resume Screening
 - Shortlisted based on Python skills, ML project quality, and GitHub activity
- Round 2 – Online Assessment (60 mins)
 - Python MCQs, one data manipulation problem (Pandas/NumPy), one ML conceptual question
- Round 3 – Technical Interview (30–45 mins)

- Discussion of your past projects (especially any RAG / ML pipeline work)
- Basic ML concept questions — logistic regression, evaluation metrics, overfitting
- Live Python problem-solving — small data cleaning or model evaluation task
- Round 4 – HR Discussion (15 mins)
 - Cultural fit, start date, stipend discussion

HOW TO APPLY

Send your resume + GitHub profile link to:

careers@techspark.in

Subject line: **ML Intern Application – [Your Name] – [College Name]**

Applications reviewed on a rolling basis. Early applicants are given preference.